Matrix

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Matrix

Note:

(i) In a compact form, the matrix (as shown in definition) can also be represented as $A = [a_{ij}]$, $1 \le i \le m$, $1 \le j \le n$ or simply $[a_{ij}]_{m\times n}$.

(ii) a_{11}, a_{12}, ...,(numbers in the matix) are called the elements of the matrix.

(iii) Any element that lies in the ith row and j^{th} column of a matrix is represented by a_{ij} and it is called the $(i, j)^{th}$ element of the matrix.

Examples

(i) $A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 3 & 2 \end{bmatrix}$ This matrix has two rows and 3 columns so it's order will be 2×3 and it's elements can be represented as $a_{11} = 2$, $a_{12} = 1$, $a_{13} = -1$, $a_{21} = 1$, $a_{22} = 3$, $a_{23} = 2$.

(ii) $B = \begin{bmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{bmatrix}$ iThis matrix has 2 rows and 2 columns so it's order will be 2×2 and it's elements can be represented as $b_{11} = \sin x$, $b_{12} = \cos x$, $b_{21} = \cos x$, $b_{22} = -\sin x$.

Definition

A rectangular array of real or complex numbers (symbols) arranged in row and column is called matrix. A matrix of order m×n means it will have m rows and n columns. The total number of elements in this matrix will be equal to mn. A matrix of order m×n can be represented as:

(Matrices are represented by [ ], ( ))

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Q. **What are the possible orders a matrix can have, if it has 12 elements?**

Sol. Let matrix has order m × n, then mn = 12
So, possible orders are 1 × 12, 2 × 6, 3 × 4, 4 × 3, 6 × 2, 12 × 1

Q. **Construct the matrix of order 3 × 2 whose elements are given by $a_{ij} = 2i - j$.**

Sol. $A = \begin{bmatrix} 1 & 0 \\ 3 & 2 \\ 5 & 4 \end{bmatrix}$

TYPES OF MATRICES

Row Matrix

If in a matrix the number of row is one, then it is called row matrix.

Example :

A = [10 13 −21 −12] is a row matrix of order 1 × 4.

Column matrix

If in a matrix the number of column is one, then it is called column matrix or column vector.

Example:

$A = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 \\ 2 \\ 5 \\ 4 \end{bmatrix}$ are column-matrix of

order 3 × 1 and 4 × 1 respectively.

Square matrix

If in a matrix the number of rows and columns are equal then this matrix is called square matrix. This matrix will be called of order n if it it's number of rows and columns are n.

Example :

The matrix $\begin{bmatrix} 2 & 1 & -1 \\ 3 & -2 & 5 \\ 1 & 5 & -3 \end{bmatrix}$ is square matrix of

order 3 in which the diagonal elements are 2, −2 and −3.

Diagonal matrix

If all the elements of a matrix except those which are in the leading diagonal is zero, then this matrix is called diagonal matrix, i.e., $a_{ij} = 0$, for all $i \neq j$

Point to Remember!!!

The elements a_{ij} in a square matrix $A = [a_{ij}]_{r \times n}$ having the property i = j, i.e., the elements $a_{11}, a_{22}, ..., a_{nn}$ are called the Diagonal elements. The line along which these elements lie is called the Principal Diagonal or Leading Diagonal of the matrix.

Know the facts



If $d_1, d_2, ..., d_n$ are the diagonal elements of a n × n diagonal matrix, then this matrix is denoted by diag $[d_1, d_2, ..., d_n]$.

Example :

The matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ is a diagonal matrix,

It can be represented as A = diag (1, 2, 3).

Know the facts

A diagonal matrix is said to be a scalar matrix if all of it's diagonal elements are equal.

Scalar Matrix

A square matrix $A = [a_{ij}]_{n\times n}$ is known as Scalar Matrix if

(i) $a_{ij} = 0$ for all $i \neq j$, and

(ii) $a_{ii} = c$ for all i, where $c \neq 0$.

Example :

The matrices $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ and

$$B = \begin{bmatrix} 1-2i & 0 & 0 \\ 0 & 1-2i & 0 \\ 0 & 0 & 1-2i \end{bmatrix}$$

Above matrices are scalar matrices of order 2 and 3 respectively.

Identity or Unit Matrix

Know the facts

The identity matrix of order n is denoted by I_n.

A square matrix $A = [a_{ij}]_{n\times n}$ is called an identity or unit matrix if

(i) $a_{ij} = 0$ for all $i \neq j$ and,

(ii) $a_{ii} = 1$ for all i

A diagonal matrix is said to be a identity matrix if all of it's diagonal elements are equal to 1.

Example:

The matrices $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ are

identity matrices of order 2 and 3 respectively.

Null matrix

If all the elements of a matrix is equal to zero then it called null matrix or void matrix.

Example:

$\begin{bmatrix}0 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}0 & 0 & 0\\0 & 0 & 0\end{bmatrix}$ are null matrices of order 2 × 2 and 2 × 3 respectively.

Real Matrix

If all the elements of a matrix if real then it is called real matrix.

Complex Matrix

If at least one element of a matrix is an imaginary number then this matrix is called a complex matrix..

Sub Matrix

If we leave some rows and columns of a m × n matrix A, then the matrix thus obtained is called sub matrix of A.

Example :

$$A = \begin{bmatrix}2 & 2 & 5 & 6\\5 & 7 & 1 & 0\\2 & 1 & 1 & 7\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}\cancel{2} & \cancel{2} & \cancel{5} & \cancel{6}\\5 & 7 & \cancel{1} & 0\\2 & 1 & \cancel{1} & 7\end{bmatrix} = \text{(sub matrix of A)} = \begin{bmatrix}5 & 7 & 0\\2 & 1 & 7\end{bmatrix}$$

Also, $\begin{bmatrix}\cancel{2} & \cancel{2} & \cancel{5} & \cancel{6}\\\cancel{5} & \cancel{7} & \cancel{1} & \cancel{0}\\2 & 1 & \cancel{1} & 7\end{bmatrix}$ = (sub matrix of A) = [2 1 7]

Singular Matrix

If $|A| \neq 0$, then the square matix is called non-singular square matrix, if $|A| = 0$, then it is called square singular matrix (where $|A|$ denotes the determinant of matrix A)

EQUALITY OF MATRICES

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ can be called equal, if :

(i) They are of the same order,

(ii) Each element of 'A' is equal to the corresponding element of B, that is $a_{ij} = b_{ij}$ for all i and j.

Example :

$\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ and $\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ are equal matrices but $\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ and $\begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}$ can not be called equal matrices. Figuratively if two matrices A and B are equal, we write A = B.

Example :

If $\begin{bmatrix} x & y \\ z & a \\ b & c \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 2 & \sqrt{3} \\ \sqrt{5} & \frac{1}{2} \end{bmatrix}$, then $x = -1,\ y = 0, z = 2, a = \sqrt{3}$, $b = \sqrt{5}$, $c = \frac{1}{2}$

Q. **If $\begin{bmatrix} x+y & 2x+z \\ x-y & 2z+w \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 0 & 10 \end{bmatrix}$, then find the values of x, y, z, w.**

Sol. By equating elements

$x + y = 4$ ...(1)

$x - y = 0$ ...(2)

gives $x = 2, y = 2$

Also, $2x + z = 7 \Rightarrow z = 3$

And $2z + w = 10 \Rightarrow w = 4$.

Q. **Find the values of x and y for which the following matrices are equal.**

$$A = \begin{bmatrix} 2x+1 & 3y \\ 0 & y^2-5y \end{bmatrix},\ B = \begin{bmatrix} x+3 & y^2+2 \\ 0 & -6 \end{bmatrix}$$

Sol. By equating elements

$2x + 1 = x + 3 \Rightarrow x = 2$

$3y = y^2 + 2 \Rightarrow y = 1, 2$ ...(1)

$y^2 - 5y = -6 \Rightarrow y = 2, 3$ ...(2)

by (1) and (2) $y = 2$

So, $x = 2, y = 2$.

Q. If A is a n-rowed square matrix, $A = [a_{ij}]$ where $a_{ij} = \left[\frac{i}{j}\right]$, [×] denotes greatest integer, then find the value of det (A).

(A) 2 **(B) 0** **(C) 1** **(D) –1**

Sol. **(C)**

$$A = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 2 & 1 & 0 & \dots & 0 \\ 3 & 1 & 1 & \dots & 0 \\ \vdots & & & & \\ n & \dots\dots\dots\dots & & & 1 \end{bmatrix}$$

$|A| = 1 \times 1 \times 1 \times \dots x1$ (n times) $+ 0 + 0 + 0 \dots + 0$; $|A| = 1$

Q. The number of distinct values of x for which matrix $\begin{bmatrix} 3 & x-1 & 2 \\ 3 & -1 & x+2 \\ x+3 & -1 & 2 \end{bmatrix}$ is singular, is:

(A) 2 **(B) 0** **(C) 1** **(D) 3**

Sol. **(A)**

$\Delta = 3(-2 + x + 2) - (x - 1)\{(6 - (x + 3)(x + 2)\} + 2\{-3 + x + 3\} = 0$

$\Rightarrow 3x + (x - 1)(x^2 + 5x) + 2x = 0$

$\Rightarrow x\{(x - 1)(x + 5) + 5\} = 0$

$\Rightarrow x\{x^2 + 4x\} = 0$

$\Rightarrow x^2(x + 4) = 0$

$\Rightarrow x = \{0, -4\}$

Q. A is square matrix of order n.I = maximum number of distinct entries if A is a triangular matrix m = maximum number of distinct entries if A is a diagonal matrix p = minimum number of zeroes if A is a triangular matrix If I + 5 = p + 2m, find the order of the matrix.

(A) 8 **(B) 6** **(C) 5** **(D) 4**

Sol. **(D)**

$$I = \frac{n(n+1)}{2} + 1$$

$m = n + 1$

$p = \frac{n(n-1)}{2}$

Now, $l + 5 = p + 2m$

$\frac{n(n+1)}{2} + 1 + 5 = \frac{n(n-1)}{2} + 2(n+1)$

$\Rightarrow \frac{n}{2}\{(n+1) - (n-1)\} + 6 = 2n + 2$

$\Rightarrow n + 4 = 2n$

$\Rightarrow n = 4$

Q. **If a matrix has 18 elements, find out all of it's possible order. What, if it has 5 elements?**

Sol. Let order be $m \times n$ then

(i) $mn = 18$

$(m, n) = \{(1, 18), (2, 9), (3, 6), (6, 3), (9, 2). (18, 1)\}$

(ii) $mn = 5$

$\Rightarrow m \times n = \{1 \times 5, 5 \times 1\}$

Q. **Find the number of all possible matrices of order 3 × 3 having each entry either 0 or 1.**

(A) 27 **(B) 256** **(C) 81** **(D) 512**

Sol. **(D)**

For a 3 × 3 matrix, we have 9 positions.

Each position can be filled with 0 or 1

Total number of matrices = $\underbrace{2 \times 2x...2}_{9 \text{ times}} = 2^9$

ALGEBRA OF MATRICES

Addition of matrices

(i) Any two matrices can only be added if they are of the same order. The matrix thus produced is of the same order. Hence we can say that if two matrices A and B are of the same order, only then they can be conformable for addition.

(ii) If $A = [a_{ij}]_{m\times n}$ and $B = [b_{ij}]_{m\times n}$ are two matrices of the same order, their sum A + B is defined as the matrix of order m × n such that

$(A + B)_{ij} = a_{ij} + b_{ij}$ for i = 1, 2, ..., m and j = 1, 2, ..., n.

Example :

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$ then

$$A + B = \begin{bmatrix} 1+6 & 2+5 & 3+4 \\ 4+3 & 5+2 & 6+1 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 7 & 7 & 7 \\ 7 & 7 & 7 \end{bmatrix}$$

Example :

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 & 1 \\ 3 & 2 & 1 \\ 2 & 5 & -2 \end{bmatrix}$,

Since A and B are not of the same order, A + B is not defined.

Properties of Matrix addition

(i) If A and B are two m × n matrices, then A + B = B + A, it means matrix addition is commutative

(ii) If A, B, C are three matrices of the same order, then (A + B) + C = A + (B + C). it means matrix addition is associative

(iii) The null matrix is the identity element for matrix addition, i.e., A + O = A = O + A, it means identity element exists for matrix A

(iv) For every matrix $A = [a_{ij}]_{m \times n}$ there exists a matrix $[-a_{ij}]_{m \times n}$, denoted by −A, such that A + (−A) = O = (−A) + A, it means inverse exists for matrix A.

(v) Cancellation laws hold true in case of addition of matrices If A, B, C are matrices of the same order, then

A + B = A + C ⇒ B = C (left cancellation law)

and, B + A = C + A ⇒ B = C (right cancellation law)

Subtraction of Matrices

Definition :

For two matrices A and B of the same order, we define A − B = A + (−B).

Q. If $A=\begin{bmatrix}-3 & 2 & 1\\ 1 & -4 & 7\end{bmatrix}$ and $B=\begin{bmatrix}3 & 5 & -2\\ -1 & 4 & -2\end{bmatrix}$, then A − B = A + (−B)

Sol. $A-B=\begin{bmatrix}-3 & 2 & 1\\ 1 & -4 & 7\end{bmatrix}-\begin{bmatrix}3 & 5 & -2\\ -1 & 4 & 2\end{bmatrix}=\begin{bmatrix}-3-3 & 2-5 & 1-(-2)\\ 1-(-1) & -4-4 & 7-(-2)\end{bmatrix}=\begin{bmatrix}-6 & -3 & 3\\ 2 & -8 & 9\end{bmatrix}$

$A+B=\begin{bmatrix}-3 & 2 & 1\\ 1 & -4 & 7\end{bmatrix}+\begin{bmatrix}3 & 5 & -2\\ 1 & 4 & 2\end{bmatrix}=\begin{bmatrix}-3-3 & 2-5 & 1+2\\ 1+2 & -4-4 & 7+2\end{bmatrix}=\begin{bmatrix}-6 & -3 & 3\\ 2 & -8 & 9\end{bmatrix}$

Clearly, both LHS and RHS are equal

Multiplication of Matrix by scalar

If $A=\begin{bmatrix}2 & 3 & 5\\ 6 & 7 & 8\end{bmatrix}_{2\times3}$, then $2A=\begin{bmatrix}4 & 6 & 10\\ 12 & 14 & 16\end{bmatrix}_{2\times3}$

Properties of Scalar Multiplication

If $A=[a_{ij}]_{m\times n}$, $B=[b_{ij}]_{m\times n}$ are two matrices and k, l are scalars, then

(i) $k(A+B)=kA+kB$

(ii) $(k+l)A=kA+lA$

(iii) $(kl)A=k(lA)=l(kA)$

(iv) $(-k)A=-(kA)=k(-A)$

(v) $1.A=A$

(vi) $(-1)A=-A$.

Q. If $A=\begin{bmatrix}2 & -1\\ 3 & 1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 4\\ 7 & 2\end{bmatrix}$, find 3A − 2B.

Sol. $A=\begin{bmatrix}2 & -1\\ 3 & 1\end{bmatrix}\Rightarrow 3A=\begin{bmatrix}6 & -3\\ 9 & 3\end{bmatrix}$

$B=\begin{bmatrix}1 & 4\\ 7 & 2\end{bmatrix}\Rightarrow 2B=\begin{bmatrix}2 & 8\\ 14 & 4\end{bmatrix}$

$3A-2B=\begin{bmatrix}6-2 & -3-8\\ 9-14 & 3-4\end{bmatrix}=\begin{bmatrix}4 & -11\\ -5 & -1\end{bmatrix}$

Q. If $A\begin{bmatrix}2 & 3 & 4\\ 0 & 4 & 6\\ 5 & 8 & 9\end{bmatrix}, B=\begin{bmatrix}3 & 0 & 15\\ 5 & 3 & 2\\ 0 & 4 & 7\end{bmatrix}$, find 3A – 2B.

Sol. $A=\begin{bmatrix}2 & 3 & 4\\ 0 & 4 & 6\\ 5 & 8 & 9\end{bmatrix} \Rightarrow 3A=\begin{bmatrix}6 & 9 & 12\\ 0 & 12 & 18\\ 15 & 24 & 27\end{bmatrix}$

$B=\begin{bmatrix}3 & 0 & 15\\ 5 & 3 & 2\\ 0 & 4 & 7\end{bmatrix} \Rightarrow 2B=\begin{bmatrix}6 & 0 & 30\\ 10 & 6 & 4\\ 0 & 8 & 14\end{bmatrix}$

$3A-2B=\begin{bmatrix}6 & 9 & 12\\ 0 & 12 & 18\\ 15 & 24 & 27\end{bmatrix}-\begin{bmatrix}6 & 0 & 30\\ 10 & 6 & 4\\ 0 & 8 & 14\end{bmatrix}=\begin{bmatrix}0 & 9 & -18\\ -10 & 6 & 14\\ 15 & 16 & 13\end{bmatrix}$

Q. If A = diag (1, – 1, 2) and B = diag (2, 3, – 1), then find 3 A + 4B.

Sol. $A=\begin{bmatrix}1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & 2\end{bmatrix} \Rightarrow 3A=\begin{bmatrix}3 & 0 & 0\\ 0 & -3 & 0\\ 0 & 0 & 6\end{bmatrix}$

$B=\begin{bmatrix}2 & 0 & 0\\ 0 & 3 & 0\\ 0 & 0 & -1\end{bmatrix} \Rightarrow 4B=\begin{bmatrix}8 & 0 & 0\\ 0 & 12 & 0\\ 0 & 0 & -4\end{bmatrix}$

$3A+4B=\begin{bmatrix}11 & 0 & 0\\ 0 & 9 & 0\\ 0 & 0 & 2\end{bmatrix}$ = diag (11, 9, 2)

Q. If $A\ A=\begin{bmatrix}1 & 0\\ 3 & -1\\ -5 & 2\end{bmatrix}$ and $B=\begin{bmatrix}1 & -2\\ -2 & 2\\ 1 & 1\end{bmatrix}$, then find the matrix 'X' such that 3A + 2X = 5 B Q.

Sol. $3A=\begin{bmatrix}3 & 0\\ 9 & -3\\ -15 & 6\end{bmatrix}, 5B=\begin{bmatrix}5 & -10\\ -10 & 10\\ 5 & 5\end{bmatrix}$

$$5B - 3A = \begin{bmatrix} 2 & -10 \\ -19 & 13 \\ 20 & -1 \end{bmatrix}$$

$$3A + 2X = 5B \Rightarrow X = \frac{5B - 3A}{2}$$

$$X = \frac{1}{2}\begin{bmatrix} 2 & -10 \\ -19 & 13 \\ 20 & -1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & -5 \\ -\frac{19}{2} & \frac{13}{2} \\ 10 & -\frac{1}{2} \end{bmatrix}$$

Q. **Find X and Y, if $X + Y = \begin{bmatrix} 5 & 2 \\ 0 & 9 \end{bmatrix}$ and $X - Y = \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix}$**

Sol. Adding $2X = \begin{bmatrix} 5+3 & 2+6 \\ 0+0 & 9-1 \end{bmatrix} = \begin{bmatrix} 8 & 8 \\ 0 & 8 \end{bmatrix} \Rightarrow X = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix}$

Subtracting $2Y = \begin{bmatrix} 5-3 & 2-6 \\ 0-0 & 9-(-1) \end{bmatrix} = \begin{bmatrix} 2 & -4 \\ 0 & 10 \end{bmatrix}$

$$\Rightarrow \quad y = \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix}$$

Q. **If $A = \begin{bmatrix} 9 & 1 \\ 7 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 \\ 7 & 12 \end{bmatrix}$, then the matrix C such that 5A + 3B + 2C is a null matrix, is:**

(A) $\begin{bmatrix} -24 & -10 \\ -28 & -38 \end{bmatrix}$

(B) $\begin{bmatrix} -24 & -20 \\ -14 & -38 \end{bmatrix}$

(C) $\begin{bmatrix} -48 & -20 \\ -56 & -76 \end{bmatrix}$

(D) None of these

Sol. **(A)**

$$5\begin{bmatrix}9 & 1\\ 7 & 8\end{bmatrix}+3\begin{bmatrix}1 & 5\\ 7 & 12\end{bmatrix}+2C=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}45 & 5\\ 35 & 40\end{bmatrix}+\begin{bmatrix}3 & 15\\ 21 & 36\end{bmatrix}+2C=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$$\Rightarrow \begin{bmatrix}48 & 20\\ 56 & 76\end{bmatrix}+2C=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$$

$$\Rightarrow 2C=\begin{bmatrix}-48 & -20\\ -56 & -76\end{bmatrix}$$

$$\Rightarrow C=\begin{bmatrix}-24 & -10\\ -28 & -38\end{bmatrix}$$

Q. **Find the value of λ, a non-zero scalar, if**

$$\lambda\begin{bmatrix}1 & 0 & 2\\ 3 & 4 & 3\end{bmatrix}+2\begin{bmatrix}1 & 2 & 3\\ -1 & -3 & 2\end{bmatrix}=\begin{bmatrix}4 & 4 & 10\\ 4 & 2 & 10\end{bmatrix}$$

Sol.

$$\begin{bmatrix}\lambda & 0 & 2\lambda\\ 3\lambda & 4\lambda & 3\lambda\end{bmatrix}=\begin{bmatrix}4 & 4 & 10\\ 4 & 2 & 10\end{bmatrix}-\begin{bmatrix}2 & 4 & 6\\ -2 & -6 & 4\end{bmatrix}=\begin{bmatrix}2 & 0 & 4\\ 6 & 8 & 6\end{bmatrix}$$

$\lambda = 2$

Q. **Solve the matrix equation** $\begin{bmatrix}x^2\\ y^2\end{bmatrix}-3\begin{bmatrix}x\\ 2y\end{bmatrix}=\begin{bmatrix}-2\\ -9\end{bmatrix}$.

Sol.

$$\begin{bmatrix}x^2\\ y^2\end{bmatrix}-\begin{bmatrix}3x\\ 6y\end{bmatrix}=\begin{bmatrix}-2\\ -9\end{bmatrix}$$

$$\begin{bmatrix}x^2-3x\\ y^2-6y\end{bmatrix}=\begin{bmatrix}-2\\ -9\end{bmatrix}$$

$x^2 - 3x = -2$ and $y^2 - 6y = -9$

$x^2 - 3x + 2 = 0$ and $y^2 - 6y + 9 = 0$

$x = 1, 2$ and $y = 3$

$\Rightarrow (x, y) = \{(1, 3), (2, 3)\}$

Q. **Simplify** $\cos\theta\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}+\sin\theta\begin{bmatrix}\sin\theta & -\cos\theta\\ \cos\theta & \sin\theta\end{bmatrix}$

Sol.

$$\begin{bmatrix}\cos^2\theta & \cos\theta\sin\theta\\ -\cos\theta\sin\theta & \cos^2\theta\end{bmatrix}+\begin{bmatrix}\sin^2\theta & -\sin\theta\cos\theta\\ \sin\theta\cos\theta & \sin^2\theta\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=I_2$$

Trace of a Matrix

The sum of all the elements of the principal diagonal of a square matrix A is called the trace of matrix A, it is denoted by tr (A).

Thus if $A = [a_{ij}]_{n\times n}$, then

$$\text{tr}(A) = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + \ldots + a_{nn}$$

Properties of trace of a matrix

Let $A = [a_{ij}]_{n\times n}$ and $B = [b_{ij}]_{n\times n}$ and λ be a scalar, then

(i) $\text{tr}(\lambda A) = \lambda\ \text{tr}(A)$

(ii) $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$

(iii) $\text{tr}(AB) = \text{tr}(BA)$

Q. **Let** $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ **and** $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$**, then find tr (A) – tr (B).**

(A) 2 **(B) 0** **(C) 1** **(D) –1**

Sol. **(A)**

$\text{tr}(A + 2B) = 1 + (-3) + 1$

$\Rightarrow$ $\text{tr}(A) + 2\,\text{tr}(B) = -1$ ...(1)

and $\text{tr}(2A - B) = 2 + (-1) + 2$

$\Rightarrow$ $2\text{tr}(A) - \text{tr}(B) = 3$ ...(2)

By solving (1) and (2)

$\text{tr}(A) = 1,\ \text{tr}(B) = -1$

$\Rightarrow \text{tr}(A) - \text{tr}(B) = 2$

Multiplication of Matrices

Example : (i) $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$

(ii) $\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} ax+by+cz \\ dx+ey+fz \end{bmatrix}$

Note :

(i) Two matrices A and B can be multiplied only if the number of columns in A (pre-multiplier) is equal to the number of rows in B (post-multiplier).

(ii) Thus, if A is a matrix of order m × n i.e. $A = [a_{ij}]_{m\times n}$ and B is a matrix of order m × p i.e. $B = [b_{ij}]_{n\times p}$ then their product AB is of order m × p

Q. **If** $A = \begin{bmatrix} 2 & 1 & 3 \\ 3 & -2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$ **and** $B = \begin{bmatrix} 1 & -2 \\ 2 & 1 \\ 1 & -3 \end{bmatrix}$**, then find AB if possible.**

Sol.
$$AB = \begin{bmatrix} 2\times1+1\times2+3\times1 & 2\times(-2)+1\times1+3x(-3) \\ 3\times1+(-2)\times2+1\times1 & 3\times(-2)+(-2)\times1+1\times(-3) \\ -1\times1+0\times2+1\times1 & -1\times(-2)+0\times1+1\times(-3) \end{bmatrix}$$

$$AB = \begin{bmatrix} 7 & -12 \\ 0 & -11 \\ 0 & -1 \end{bmatrix}$$

Q. **Let** $A = \begin{bmatrix} 1 & -2 & 3 \\ 3 & 2 & -1 \end{bmatrix}$ **and** $B = \begin{bmatrix} 2 & 3 \\ -1 & 2 \\ 2 & -5 \end{bmatrix}$**. Find AB and BA and show that AB ≠ BA.**

Sol.
$$AB = \begin{bmatrix} 1\times2+(-2)\times(-1)+3\times2 & 1\times3+(-2)\times2+3\times(-5) \\ 3\times2+2x(-1)+(-1)\times2 & 3\times3+2\times2+(-1)\times(-5) \end{bmatrix}$$

$$AB = \begin{bmatrix} 10 & -16 \\ 2 & 18 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2\times1+3\times3 & 2(-2)+3\times2 & 2\times3+3\times(-1) \\ (-1)\times1+2\times3 & (-1)(-2)+2\times2 & (-1)3+2(-1) \\ 2\times1+(-5)\times3 & 2(-2)+(-5)\times2 & 2\times3+(-5)(-1) \end{bmatrix}$$

$$= \begin{bmatrix} 11 & 2 & 3 \\ 5 & 6 & -5 \\ -13 & -14 & 11 \end{bmatrix}$$

Clearly, AB ≠ BA

Properties of matrix multiplication

(i) Commutative law does not necessarily hold for matrix multiplication.

(ii) If AB = BA, then matrices A and B are called commutative matrices.

(iii) If AB = −BA, then matrices A and B are called anticommutative matrices.

(iv) Matrix multiplication is associative : A (BC) = (AB) C.

(v) Matrix multiplication is distributive with respect to addition
A (B ± C) = AB ± AC.

(vi) If the product AB = O, it is not necessary that at least one of the matrices should be a zero matrix.

Example :

If $\mathbf{A} = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. then $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, while neither A nor B is the null matrix.

(vii) Cancellation law does not necessarily hold, i.e., if AB = AC, then in general B ≠ C, even if A ≠ O.

(viii) If A = diag (a_1, a_2, a_3, ..., a_n) and B = diag (b_1, b_2, b_3, ..., b_n), then

$A \times B$ = diag (a_1b_1, a_2b_2, ..., a_nb_n).

Thus, A^r = diag $(a_1^n, a_2^n, a_3^n, ..., a_n^n)$

(ix) If A and B are diagonal matrices of the same order, then AB = BA or diagonal matrices are commutative.

(x) If A and B are commutative, then

$(A + B)^2 = (A + B)(A + B) = A^2 + AB + BA + B^2 = A^2 + 2AB + B^2$

Similarly

$(A + B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$

In general,

$(A + B)^n = {}^nC_0A^n + {}^nC_1A^{r-1}B + {}^rC_2A^{n-2}B^2 + \cdots + {}^nC_nB^n$

Matrices A and I are always commutative. Hence,

$(I + A)^r = {}^nC_0 + {}^nC_1A + {}^rC_1A^2 + \cdots + {}^nC_rA^n$

(xi) If A_1, A_2, ..., A_n are square matrices of the same order, then $|A_1A_2 ...A_n| = |A_1||A_2| ... |A_n|$.

(xii) If k is scalar, then $|kA| = k^n|A|$, where n is order of the matrix A.

(xiii) If A and B are square matrices of same order, then |AB| = |BA| even

though AB ≠ BA.

Q. **Find the value of x and y that satisfy the equations** $\begin{bmatrix} 3 & -2 \\ 3 & 0 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} y & y \\ x & x \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3y & 3y \\ 10 & 10 \end{bmatrix}$.

Sol. $\begin{bmatrix} 3 & -2 \\ 3 & 0 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} y & y \\ x & x \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3y & 3y \\ 10 & 10 \end{bmatrix}$

$\begin{bmatrix} 3y-2x & 3y-2x \\ 3y & 3y \\ 2y+4x & 2y+4x \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3y & 3y \\ 10 & 10 \end{bmatrix}$

$3y - 2x = 3,\ 2y + 4x = 10$

$y + 2x = 5$

$y = 2,\ x = 3/2$

Q. **Let** $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ **and** $B = \begin{bmatrix} p \\ q \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ **are such that AB = B and a + d = 2, then find the value of (ad – bc).**

(A) 2 **(B) 0** **(C) 1** **(D) –1**

Sol. **(C)**

$AB = B$

$\Rightarrow \begin{bmatrix} ap+bq \\ cp+dq \end{bmatrix} = \begin{bmatrix} p \\ q \end{bmatrix}$

$\Rightarrow ap + bq = p$

$cp + dq = q$

$\Rightarrow p(a - 1) + qb = 0$

$pc + q(d - 1) = 0$

As $p, q \neq 0, 0$

So, $\frac{a-1}{c} = \frac{b}{d-1}$

$\Rightarrow (a - 1)(d - 1) = bc$

$\Rightarrow ad - bc = a + d - 1$

$\Rightarrow ad - bc = 2 - 1 = 1$

Q. **Prove that the product of matrices**

$\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}$ **and** $\begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$ **is the null matrix,**

when θ and ϕ differ by an odd multiple of $\pi/2$.

Sol. $\cos\theta\cos\phi + \sin\theta\sin\phi = \cos(\theta - \phi)$

$$\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}\begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta\cos\phi(\cos\theta\cos\phi + \sin\theta\sin\phi) & \cos\theta\sin\phi(\cos\theta\cos\phi + \sin\theta\sin\phi) \\ \sin\theta\cos\phi(\cos\theta\cos\phi + \sin\theta\sin\phi) & \sin\theta\sin\phi(\cos\theta\cos\phi + \sin\theta\sin\phi) \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta\cos\phi\cos(\theta-\phi) & \cos\theta\sin\phi\cos(\theta-\phi) \\ \sin\theta\cos\phi\cos(\theta-\phi) & \sin\theta\sin\phi\cos(\theta-\phi) \end{bmatrix}$$

Given $\theta - \phi = (2n+1)\dfrac{\pi}{2} \Rightarrow \cos(\theta - \phi) = 0$

therefore $\begin{bmatrix} \cos\theta\cos\phi\cos(\theta-\phi) & \cos\theta\sin\phi\cos(\theta-\phi) \\ \sin\theta\cos\phi\cos(\theta-\phi) & \sin\theta\sin\phi\cos(\theta-\phi) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Q. **Find all matrices which commute with A =** $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Sol. Let B be matrix such that A & B are commute

$\Rightarrow$ AB = BA

$(2 \times 2)\ (m \times n)$ $(m \times n)\ (2 \times 2)$

$\Rightarrow$ m = n = 2. So, Let $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Now, $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} a+c & b+d \\ c & d \end{bmatrix} = \begin{bmatrix} a & a+b \\ c & c+d \end{bmatrix}$

equating elements

$a + c = a \Rightarrow c = 0$

$b + d = a + b \Rightarrow a = d$

$B = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}$

Q. **If the matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (a, b, c, d not all simultaneously zero) commute, find the value of $\frac{d-b}{a+c-b}$. Also show that the matrix which commutes with A is of the form $\begin{bmatrix} \alpha - \beta & 2\beta/3 \\ \beta & \alpha \end{bmatrix}$**

Sol. Given AB = BA

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} a+2c & b+2d \\ 3a+4c & 3b+4d \end{bmatrix} = \begin{bmatrix} a+3b & 2a+4b \\ c+3d & 2c+4d \end{bmatrix}$$

$a + 2c = a + 3b,\ b + 2d = 2a + 4b,\ 3a + 4c = c + 3d,\ 3b + 4d = 2c + 4d$

$2c = 3b,\ 3b + 2a = 2d,\ a + c = d,\ 3b = 2c$

If $c = \beta,\ d = \alpha$

$b = \frac{2\beta}{3},\ a = \alpha - \beta$

$$\text{Now } \frac{d-b}{a+c-b} = \frac{\alpha - \frac{2\beta}{3}}{\alpha - \beta + \beta - \frac{2\beta}{3}} = \frac{\alpha - \frac{2\beta}{3}}{\alpha - \frac{2\beta}{3}} = 1$$

POSITIVE INTEGRAL POWERS OF A SQUARE MATRIX

Suppose A is any square matrix, then we see how integral powers of A works in the following points

(i) $A^1 = A$ and,

(ii) $A^{n+1} = A^n \cdot A$, where $n \in N$.

from the above two points, we can see that:

$A^2 = AA,\ A^3 = A^2 A = (AA)\ A$. etc.

Using the above properties these can be esily seen:

(i) $A^m . A^r = A^{m+n}$ and

(ii) $(A^m)^n = A^{mn}$ for all $m, n \in N$.

MATRIX POLYNOMIAL

Suppose $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \ldots + a_{n-1} x + a_n$
is a polynomial of degree n and A is any square matrix of order n. Then,
$f(A) = a_0 A^n + a_1 A^{n-1} + a_2 A^{n-2} + \ldots + a_{n-1} A + a_n I_n$
is called a matrix polynomial.

Example:

If $f(x) = x^2 - 5x + 3$ is a polynomial and A is a square matrix, then $A^2 - 5A + 3I$ is a matrix polynomial.

Q. **If $A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$ and $(A + B)^2 = A^2 + B^2$, then (a, b) equals**

(A) (3, 2) **(B) (0, 5)** **(C) (1, 3)** **(D) (–1, 2)**

Sol. **(C)**

$$AB = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} = \begin{bmatrix} a-b & 1+1 \\ a-b & 1+1 \end{bmatrix} = \begin{bmatrix} a-b & 2 \\ a-b & 2 \end{bmatrix}$$

$$BA = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} a+1 & -(a+1) \\ b-1 & -b+1 \end{bmatrix}$$

Now, $(A + B)^2 = A^2 + B^2$

$(A + B)(A + B) = A^2 + B^2 \Rightarrow A^2 + AB + BA + B^2 = A^2 + B^2$

$\Rightarrow AB + AB = 0$

$$\Rightarrow \begin{bmatrix} a-b & 2 \\ a-b & 2 \end{bmatrix} + \begin{bmatrix} a+1 & -(a+1) \\ b-1 & -b+1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2a-b+1 & 1-a \\ a-1 & 3-b \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow 1 - a = 0,\ 3 - b = 0$$

$\Rightarrow a = 1, b = 3$

Q. **If $AB = B$ and $BA = A$ then $A^2 + B^2$ is not equal to:**

(A) AB + BA **(B) 2AB** **(C) A + B** **(D) AB(A + B)**

Sol. **(C)**

$A^2 = AA$
$= A(BA)$
$= (AB)A$
$= (B)A$
$= A = BA$

Similarly $B^2 = B = AB$

Also, $AB(A + B) = A(BA) + (AB)B = AA + BB = A + B$

Q. **If $A = \begin{bmatrix} p & q \\ 0 & 1 \end{bmatrix}$, then show that $A^8 = \begin{bmatrix} p^8 & q\left(\frac{p^8 - 1}{p - 1}\right) \\ 0 & 1 \end{bmatrix}$**

Sol. $A = \begin{bmatrix} p & q \\ 0 & 1 \end{bmatrix}$

$$A^2 = \begin{bmatrix} p & q \\ 0 & 1 \end{bmatrix}\begin{bmatrix} p & q \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} p^2 & pq + q \\ 0 & 1 \end{bmatrix}$$

$$A^3 = A^2.A = \begin{bmatrix} p^2 & pq + q \\ 0 & 1 \end{bmatrix}\begin{bmatrix} p & q \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} p^3 & p^2q + pq + q \\ 0 & 1 \end{bmatrix}$$

By observation, $A^8 = \begin{bmatrix} p^8 & q(p^7 + p^6 + \ldots + 1) \\ 0 & 1 \end{bmatrix}$

$$= \begin{bmatrix} p^8 & q\left(\frac{p^8 - 1}{p - 1}\right) \\ 0 & 1 \end{bmatrix}$$

Q. **Let $A = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$. If $(A^8 + A^6 + A^4 + A^2 + I)V = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$ then vertical vector 11V is:(where I is the 2 × 2 identity matrix)**

(A) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ **(B)** $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ **(C)** $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ **(D)** $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$

Sol. **(A)**

$$A^2 = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = 3I$$

$A^4 = (A^2)^2 = 9I$, $A^6 = (A^2)^3 = 27I$, $A^8 = (A^4)^2 = 81I$

$$(A^8 + A^6 + A^4 + A^2 + I)\, V = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$$

$$\begin{bmatrix} 121 & 0 \\ 0 & 121 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$$

$$\begin{bmatrix} 121a \\ 121b \end{bmatrix} = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$$

$121a = 0,\ 121b = 11$

$a = 0 \quad b = \frac{1}{11}$

Transpose of Matrix

Let us consider a matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3\times3} \begin{matrix} \leftarrow R_1 \\ \leftarrow R_2 \\ \leftarrow R_3 \end{matrix}$$

$$\begin{matrix} \uparrow & \uparrow & \uparrow \\ C_1 & C_2 & C_3 \end{matrix}$$

We interchange all elements of now one (R_1) with column one (C_1) and similarly row two (R_2) with column two (C_2). Finally, we inter change all the elements of row three (R_3) with all the elements of column three (C_3).

$$\text{New matrix} = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}_{3\times3}$$

This new matrix is given the name A^T or transpose of A.
Similarly, take up another example.

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}_{3\times2} \begin{matrix} \leftarrow R_1 \\ \leftarrow R_2 \\ \leftarrow R_3 \end{matrix}$$

$$\begin{matrix} \uparrow & \uparrow \\ C_1 & C_2 \end{matrix}$$

Thus, $A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}_{2\times3}$ Transpose of matrix A

Example :

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3\times2}, \text{ then } A' = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}_{2\times3}$$

Change row into column and vice-versa.

Properties of transpose of matrix

(i) Transpose of transpose of a matrix A is equal to matrix A.
$(A^T)^T = A$

$$A=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\end{bmatrix}_{2\times3}$$

$$A^T=\begin{bmatrix}a_{11} & a_{21}\\ a_{12} & a_{22}\\ a_{13} & a_{23}\end{bmatrix}_{3\times2}$$

$$(A^T)^T=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\end{bmatrix}_{2\times3}=A$$

(ii) $(kA)^T = kA^T$ (where k is any scalar.)

(iii) $(A + B)^T = A^T + B^T$

(iv) $(AB)^T = B^T A^T$; $(ABC)^T = C^T B^T A^T$

(v) $(A^n)^T = (A^T)^n$

Q. **If** $A=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$ **then find the value of** θ **satisfying the equation** $A^T + A = I_2$.

Sol. $A^T + A = I_2$

$$\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}+\begin{bmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$$\begin{bmatrix}2\cos\theta & 0\\ 0 & 2\cos\theta\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$2\cos\theta = 1$

$$\theta = 2n\pi \pm \frac{\pi}{3}$$

Q. **If** $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ a & 2 & b\end{bmatrix}$ **is a matrix satisfying** $AA^T = 9I_3$, **then find the value of a-5b=**

(A) –1 **(B) 1** **(C) 2** **(D) 3**

Sol. **(D)**

$AA^T = 9I_3$

$$\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ a & 2 & b\end{bmatrix}\begin{bmatrix}1 & 2 & a\\ 2 & 1 & 2\\ 2 & -2 & b\end{bmatrix}=9I$$

$$\begin{bmatrix} 9 & 0 & a+2b+4 \\ 0 & 9 & 2(a-b+1) \\ a+2b+4 & 2(a-b+1) & a^2+b^2+4 \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$\Rightarrow$ $a+2b+4=0$ ...(1)

$2(a-b+1)=0$ ...(2)

$a^2 + b^2 + 4 = 9$ ...(3)

By solving

$(a, b) = (-2, -1)$

$a - 5b = 3$

Q. **Consider the two matrices A and B where $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$; $B = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$. If n(A) denotes the number of elements in A such that n(XY) = 0, when the two matrices X and Y are not conformable for multiplication. If C = (AB)(B'A); D= (B'A)(AB) then, the value of $\left(\frac{n(C)\left(|D|^2 + n(D)\right)}{n(A) - n(B)} \right)$ is:**

(A) 500 **(B) 1300** **(C) 250** **(D) 650**

Sol. **(D)**

Order of A is 2 × 2, order of B is 2 × 1

C = ABB'A,

$C = A_{2\times2}\, B_{2\times1}\, B'_{1\times2}\, A_{2\times2}$

$= A_{2\times2}(BB')_{2\times2}\, A_{2\times2}$

Order of C= 2 × 2

n(C) = 4

$AA = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix}$

D = B'AAB

$D = [5 \quad -3] \begin{bmatrix} 9 & 8 \\ 16 & 17 \end{bmatrix}\begin{bmatrix} 5 \\ -3 \end{bmatrix}$

$= [5 \quad -3] \begin{bmatrix} 21 \\ 29 \end{bmatrix} = [18]$

$$\text{now } \frac{n(c)(|D|^2 + n(D))}{n(A) - n(B)} = \frac{4(324+1)}{2}$$

$= 2 \times 325 = 650$

Symmetric and Skew Symmetric Matrices.

If the property $a_{ij} = a_{ji}$ for all i, j.holds for a matrix A then the square matrix $A = [a_{ij}]$ is called a symmetric matrix

Example :

The matrix $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 2 & 5 \\ 1 & 5 & -2 \end{bmatrix}$ is symmetric, because $a_{12} = -1 = a_{21}$, $a_{13} = 1 = a_{31}$, $a_{23} = 5 = a_{32}$.

Note :

(i) It follows from the definition of a symmetric matrix that A is symmetric $\Leftrightarrow a_{ij} = a_{ji}$ for all i, j $\Leftrightarrow (A)_{ij} = (A^T)_{ij}$ for all i, j $\Leftrightarrow A = A^T$.

(ii) Thus, a square matrix A is symmetric matrix if $A^T = A$.

$$A = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}, B = \begin{bmatrix} 2+i & 1 & 3 \\ 1 & 2 & 3+2i \\ 3 & 3+2i & 4 \end{bmatrix}$$

Matrices are symmetric matrices, because $A^T = A$ and $B^T = B$.

Skew-Symmetric Matrix

f the property $a_{ij} = -a_{ji}$ for all i, j.holds for a matrix A then the square matrix $A = [a_{ij}]$ is called a skew- symmetric matrix

Example :

Matrix $A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}$ is skew-symmetric because

$a_{12} = 2, a_{21} = -2 \Rightarrow a_{12} = -a_{21}$
$a_{13} = -3, a_{31} = 3 \Rightarrow a_{13} = -a_{31}$
$a_{23} = 5, a_{32} = -5 \Rightarrow a_{23} = -a_{32}$
Thus, a square matrix A is a skew-symmetric matrix if $A^T = -A$.

Example :

Matrices $A = \begin{bmatrix} 0 & 2i & 3 \\ -2i & 0 & 4 \\ -3 & -4 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & -3 & 5 \\ 3 & 0 & 2 \\ -5 & -2 & 0 \end{bmatrix}$ are skew-symmetric matrices, because $A^T = -A$ and $B^T = -B$.

Properties of Symmetric and Skew-symmetric Matrices

(i) If A is a symmetric matrix, then negative of A, scalar multiple of A, tanspose of A, inverse of A A^r, are also symmetric matrices, where $n \in N$, $k \in R$

(ii) If A is a skew-symmetric matrix, then

(a) Even power of A i.e. A^{2n} is a symmetric matrix for $n \in N$,

(b) Odd power of A i.e. A^{2n+1} is a skew-symmetric matrix for $n \in N$,

(c) Scalar multiple of A i.e. kA is also skew-symmetric matrix, where $k \in R$,.

(iii) If A, B are two symmetric matrices, then

(a) $A \pm B$, $AB + BA$ are also symmetric matrices,

(b) $AB - BA$ is a skew-symmetric matrix,

(c) AB is a symmetric matrix, when $AB = BA$.

(iv) If A, B are two skew-symmetric matrices, then

(a) $A \pm B$, $AB - BA$ are skew-symmetric matrices,

(b) $AB + BA$ is a symmetric matrix.

Q. **Express the matrix $A = \begin{bmatrix} 3 & 2 & 3 \\ 4 & 5 & 3 \\ 2 & 4 & 5 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.**

Sol. $A = \begin{bmatrix} 3 & 2 & 3 \\ 4 & 5 & 3 \\ 2 & 4 & 5 \end{bmatrix}$

$A^T = \begin{bmatrix} 3 & 4 & 2 \\ 2 & 5 & 4 \\ 3 & 3 & 5 \end{bmatrix}$

$A + A^T = \begin{bmatrix} 6 & 6 & 5 \\ 6 & 10 & 7 \\ 5 & 7 & 10 \end{bmatrix}$

$$A - A^T = \begin{bmatrix} 0 & -2 & 1 \\ 2 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2} = \begin{bmatrix} 3 & 3 & \frac{5}{2} \\ 3 & 5 & \frac{7}{2} \\ \frac{5}{2} & \frac{7}{2} & 5 \end{bmatrix} + \begin{bmatrix} 0 & -1 & \frac{1}{2} \\ 1 & 0 & \frac{-1}{2} \\ \frac{-1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

Q. **Show that positive odd integral powers of a skew-symmetric matrix are skew-symmetric and positive even integral powers of a skew-symmetric matrix are symmetric.**

Sol. Let A be skew symmetric Matrix

(i) $B = A^{2r+1}$

$B^T = (A^{2n+1})^T$

$= (A^T)^{2n+1}$ $(\because (A^n)^T = (A^T)^r)$

$= (-A)^{2n+1}$

$= (-1)^{2n+1} (A)^{2r+1}$ $(\because (\lambda A)^n = \lambda^r . A^n, \lambda \text{ is scalar})$

$= - A^{2r+1}$

$= - B$

Hence, B is skew symmetric matrix

(ii) Let $B = A^{2n}$

$B^T = (A^{2r})^T$

$= (A^T)^{2n}$ $(\because (A^n)^T = (A^T)^r)$

$= (-A)^{2n}$

$= (-1)^{2n} (A)^{2n}$ $(\because (\lambda A)^n = \lambda^r A^n, \lambda \text{ is scalar})$

$= A^{2n}$

$= B$

Hence, B is symmetric matrix

Q. **If A and B are symmetric matrices, then show that AB is symmetric iff AB = BA i.e. A and B commute.**

Sol. Let $Z = AB$

$Z^T = (AB)^T$

$= (B^T A^T)$

$= BA$

But given Z is symmetric. So, $Z = Z^T$

$\Rightarrow$ AB = BA

If AB = BA then $(AB)^T = (BA)^T$

$\Rightarrow$ $B^TA^T = A^TB^T$

$\Rightarrow$ BA = AB **Hence Proved.**

Q. **For a square matrix N of order 5. Calculate $|(N - N^T)^{2021}|$.**

Sol. Let $Z = N - N^T$ is skew symmetric matrix

$B = Z^{2021} = (N - N^T)^{2021}$ is also skew symmetric matrix.

So, B is skew symmetric matrix of odd order.

Hence $|B| = |(N - N^T)^{2021}| = 0$

Previous Year's Question

If M is a 3 × 3 matrix, where $M^T M = I$ and det (M) = 1, then prove that det (M – I) = 0 **[JEE 2004, 2 out of 60]**

Sol. $(M - I)^T = M^T - I = M^T - M^TM = M^T (I - M)$

$\Rightarrow$ $|(M - I)^T| = |M^T| |I - M|$

$\Rightarrow$ $|M - I| = |I - M|$

$\Rightarrow$ $|M - I| = 0$.

Alternate: $\det (M - I) = \det (M - I) \det (M^T) = \det (MM^T - M^T)$

$= \det (I - M)^T = - \det (M^T - I) = - \det (M - I)^T = - \det(M - I)$

$\Rightarrow$ $\det (M - I) = 0$.

Previous Year's Question

Let X and Y be two arbitrary, 3 × 3, non-zero, skew, symmetric matrices and Z be an arbitrary 3 × 3, non-zero, symmetric matrix, Then which of the following matrices is (are) skew symmetric ? **[JEE Adv. 2015]**

(A) $Y^3Z^4 - Z^4Y^3$ (B) $X^{44} + Y^{44}$ (C) $X^4Z^3 - Z^3X^4$ (D) $X^{23} + Y^{23}$

Sol. **(CD)**

$X^T = - X$, $Y^T = - Y$, $Z^T = Z$

(A) Let $P = Y^3Z^4 - Z^4Y^3$

$P^T = (Y^3Z^4)^T - (Z^4Y^3)^T$

$- Z^4Y^3 + Y^3Z^4 = P$

$\Rightarrow$ symmetric

(B) Let $P = X^{44} + Y^{44}$

$P^T = (X^{44})^T + (Y^{44})^T = P$

$\Rightarrow$ symmetric

(C) Let $P = X^4Z^3 - Z^3X^4$

$P^T = (Z^3)^T(X^4)^T - (X^4)^T(Z^3)^T$

$= Z^3X^4 - X^4Z^3 = -P$

$\Rightarrow$ skew symmetric

(D) Let $P = X^{23} + Y^{23}$

$P^T = -X^{23} - Y^{23} = -P$

$\Rightarrow$ skew symmetric

Previous Year's Question

How many 3 × 3 matrices M with entries from {0, 1, 2} are there, for which the sum of the diagonal entries of M^TM is 5 ? **[JEE Adv. 2017]**

(A) 135 (B) 198 (C) 162 (D) 126

Sol. **(A)**

Let $M = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$

$M^TM = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$

Diagonal entry $= a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2 + h^2 + i^2 = 5$

Case-I : Five (1's) and four (0's)

$^9C_5 = 126$

Case-II: One (2) and one (1)

$^9C_2 \times 2! = 72$

$\therefore$ Total = 198

Q. $\mathbf{A = \begin{bmatrix} 3 & a & -1 \\ 2 & 5 & c \\ b & 8 & 2 \end{bmatrix}}$ **is symmetric and** $\mathbf{B = \begin{bmatrix} d & 3 & a \\ b-a & e & -2b-c \\ -2 & 6 & -f \end{bmatrix}}$ **is skew-symmetric, then find AB.**

Sol. $A = \begin{bmatrix} 3 & a & -1 \\ 2 & 5 & c \\ b & 8 & 2 \end{bmatrix}$ is symmetric $\Rightarrow a = 2, b = -1, c = 8$

$$B = \begin{bmatrix} d & 3 & a \\ b-a & e & -2b-c \\ -2 & 6 & -f \end{bmatrix} \text{ is skew symmetric} \Rightarrow d = e = -f = 0$$

$$\text{Now, } AB = \begin{bmatrix} 3 & 2 & -1 \\ 2 & 5 & 8 \\ -1 & 8 & 2 \end{bmatrix} \begin{bmatrix} 0 & 3 & 2 \\ -3 & 0 & -6 \\ -2 & 6 & 0 \end{bmatrix} = \begin{bmatrix} -4 & 3 & -6 \\ -31 & 54 & -26 \\ -28 & 9 & -50 \end{bmatrix}$$

Q. **Find a Matrix X such that $X = P'Q^{2017} P$.**

Where $P = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$ **and** $Q = PAP'$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Sol. $PP' = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

$Q = PAP'$

$Q^2 = (PAP')\,(PAP')$

$= PAP'PAP' = PAIAP' = PA^2P'$

$Q^3 = PAP'PAP'PAP'$

$= PAIAIAP'$

$= P(AI)\,(AI)\,AP'$

$= PAAAP'$

$= PA^3P'$

by observation

$Q^{2017} = PA^{2017}P'$

$X = P'Q^{2017} P$

$= P'PA^{2017} P'P$

$= I(A^{2017})\, I \;= A^{2017}$

$$\text{Now, } A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$A^3 = A^2 A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$\text{by observation } A^{2017} = \begin{bmatrix} 1 & 2017 \\ 0 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 2017 \\ 0 & 1 \end{bmatrix}$$

Q. **Let** $\Delta = \begin{vmatrix} a & 4-i & 1+i \\ 4+i & b & 3+i \\ 1-i & 3-i & c \end{vmatrix}$ **, then** Δ **is always**

(A) real **(B) imaginary** **(C) zero** **(D) none of these'**

Sol. **(A)**

Given, $\Delta = \begin{vmatrix} a & 4-i & 1+i \\ 4+i & b & 3+i \\ 1-i & 3-i & c \end{vmatrix}$

$\therefore \quad \overline{\Delta} = \begin{vmatrix} a & 4+i & 1-i \\ 4-i & b & 3-i \\ 1+i & 3+i & c \end{vmatrix}$

$= \begin{vmatrix} a & 4-i & 1+i \\ 4+i & b & 3+i \\ 1-i & 3-i & c \end{vmatrix} = \Delta \quad \left(\because |A| = |A^T|\right)$

$\therefore$ Δ is real.

SPECIAL MATRICES

Orthogonal Matrix

A square matrix A is called an orthogonal matrix if the product of the matrix A and its transpose A' is an identity matrix. i.e., AA' = A'A = I.

For $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$$AA' = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{vmatrix} \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix} = I$$

Gives the following results:

1. $a_{11}^2 + a_{12}^2 + a_{13}^2 = 1$
2. $a_{21}^2 + a_{22}^2 + a_{23}^2 = 1$
3. $a_{31}^2 + a_{32}^2 + a_{33}^2 = 1$
4. $a_{11}a_{21} + a_{12}a_{22} + a_{13}a_{23} = 0$
5. $a_{21}a_{31} + a_{22}a_{32} + a_{23}a_{33} = 0$
6. $a_{31}a_{11} + a_{32}a_{12} + a_{33}a_{13} = 0$

Similarly, If we do A'A = I we get:

1. $a_{11}^2 + a_{21}^2 + a_{31}^2 = 1$
2. $a_{12}^2 + a_{22}^2 + a_{32}^2 = 1$
3. $a_{13}^2 + a_{23}^2 + a_{33}^2 = 1$
4. $a_{11}a_{12} + a_{21}a_{22} + a_{31}a_{32} = 0$
5. $a_{12}a_{13} + a_{22}a_{23} + a_{32}a_{33} = 0$
6. $a_{13}a_{11} + a_{23}a_{21} + a_{33}a_{31} = 0$

(i) We will say A' is invers of A if AA' = I
(ii) For 2 orthogonal matrices A and B, AB will also be orthogonal.
(iii) For orthogonal matrix, value of determinant = ± 1.

Idempotent Matrix

For square matrix A to be idempotent, it should satisfy relation $A^2 = A$.

Periodic Matrix

If $A^{k+1} = A$, where k is least positive integer for which this relation satisfies, then 'k' will be period of A.

Nilpotent Matrix

For square matrix A to be nilpotent of order k, A^k should beequal to '0'.
$A^k = 0$ (null matrix)
It should be noted that 'k' is a positive integer.

Involutory Matrix

For square matrix A to be involutory, $A^2 = I$ (identity matrix)

Example :

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \text{ and } A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \begin{cases} A^n = I & \text{for } n = \text{even} \\ A^n = A & \text{for } n = \text{odd} \end{cases}$$

Q. **If $i = \sqrt{-1}$, $a = \dfrac{1+\sqrt{5}}{2}$ and $b = \dfrac{1-\sqrt{5}}{2}$, then which of the following is an idempotent matrix?**

(A) $\begin{bmatrix} a & i \\ i & -b \end{bmatrix}$ **(B)** $\begin{bmatrix} b & i \\ i & -a \end{bmatrix}$ **(C)** $\begin{bmatrix} a & i \\ i & b \end{bmatrix}$ **(D)** $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$

Sol. **(C)**

For idempotent matrix, $A^2 = A$

Let $A = \begin{bmatrix} a & i \\ i & b \end{bmatrix}$, then

$$A^2 = A \cdot A = \begin{bmatrix} a & i \\ i & b \end{bmatrix}\begin{bmatrix} a & i \\ i & b \end{bmatrix} = \begin{bmatrix} a^2-1 & i(a+b) \\ i(a+b) & b^2-1 \end{bmatrix}$$

Now, $a^2 - 1 = \left(\frac{1+\sqrt{5}}{2}\right)^2 - 1 = \frac{1+\sqrt{5}}{2} = a$

$$b^2 - 1 = \left(\frac{1-\sqrt{5}}{2}\right)^2 - 1 = \frac{1-\sqrt{5}}{2} = b$$

$i(a + b) = i$

So, $A^2 = \begin{bmatrix} a^2-1 & i(a+b) \\ i(a+b) & b^2-1 \end{bmatrix} = \begin{bmatrix} a & i \\ i & b \end{bmatrix} = A$

Q. **Show that the matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is idempotent.**

Sol. $$A^2 = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

$A^2 = A$

Hence, A is idempotent.

Q. **The matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is**

(A) an involuntary matrix **(B) an idempotent matrix**

(C) a nilpotent matrix of order 3 **(D) a nilpotent matrix of order 4**

Sol. **(C)**

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1+5-6 & 1+2-3 & 3+6-9 \\ 5+10-12 & 5+4-6 & 15+12-18 \\ -2-5+6 & -2-2+3 & -6-6+9 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}$$

$$A^3 = A^2A = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$ A^3 is null matrix

Q. **Show that the matrix $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ is involutory.**

Sol. $$A^2 = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}\begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$A^2 = I$

Q. **If $(A + I)^n = I + 255A$. When A is idempotent and I is identity matrix, the value of n is:**

(A) 128 **(B) 8** **(C) 9** **(D) 10**

Sol. $A^2 = A$

$A^3 = A^2A = A.A = A^2 = A$

For all $n \in N$, $A^r = A$

Now, $(A + I)^r = {}^nC_0A^r + {}^nC_1A^{r-1}I + {}^nC_2A^{r-2}I + \ldots + {}^nC_rI^n$

$= {}^nC_0A^r + {}^nC_1A^{r-1} + {}^nC_2A^{r-2} + \ldots + {}^rC_{n-1}A + {}^nC_nI$

$= {}^rC_6A + {}^nC_1A + {}^rC_2A + \ldots + {}^nC_{n-1}A + I$

$= ({}^rC_0 + {}^nC_1 + \ldots {}^nC_{r-1})A + I$

$255A + I = (2^n - 1)A + I$

$2^n - 1 = 255$

$\Rightarrow \quad 2^n = 256 \Rightarrow \quad n = 8$

Q. **If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ and $A^n = A$ find the minimum value of n when $\theta = \pi/6$, $n \in N$**

(A) 17 **(B) 15** **(C) 13** **(D) 19**

Sol. **(C)**

$$A^2 = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\theta - \sin^2\theta & \cos\theta\sin\theta + \sin\theta\cos\theta \\ -\sin\theta\cos\theta - \cos\theta\sin\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$$

Similarly $A^3 = \begin{bmatrix} \cos 3\theta & \sin 3\theta \\ -\sin 3\theta & \cos 3\theta \end{bmatrix}$

By observation $A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}$

Now, $A^n = A$

$$\Rightarrow \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$\Rightarrow$ $\cos n\theta = \cos\theta$, $\sin n\theta = \sin\theta$

$\Rightarrow$ $n\theta = 2\pi + \theta$ (for smallest n)

$$\Rightarrow (n-1)\theta = 2\pi \Rightarrow n = \frac{2\pi}{\pi/6} + 1 \Rightarrow n = 13$$

Q. **If** $\mathbf{A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}}$ **and** $\mathbf{A^n = \begin{bmatrix} f(n) & g(n) \\ h(n) & k(n) \end{bmatrix}}$

then find the value of $\mathbf{\sum_{r=1}^{10} f(r) + \sum_{r=1}^{5} g(r) + \sum_{r=1}^{15} h(r) + \sum_{r=1}^{20} k(r)}$

Sol. $$A^2 = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 9-4 & -12+4 \\ 3-1 & -4+1 \end{bmatrix} = \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix}$$

$$A^3 = A^2A = \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix}\begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 15-8 & -20+8 \\ 6-3 & -8+3 \end{bmatrix} = \begin{bmatrix} 7 & -12 \\ 3 & -5 \end{bmatrix}$$

$$A^n = \begin{bmatrix} n^{th}\text{term of } 3,5,7,\ldots & n^{th} \text{ term of } -4,-8,-12\ldots \\ n^{th}\text{term of } 1,2,3,\ldots. & n^{th} \text{ term of } -1,-3,-5 \end{bmatrix} = \begin{bmatrix} 2n+1 & -4n \\ n & -2n+1 \end{bmatrix}$$

Now, $\sum_{r=1}^{10} f(r) + \sum_{r=1}^{5} g(r) + \sum_{r=1}^{15} h(r) + \sum_{r=1}^{20} k(r)$

$$= \sum_{r=1}^{10}(2r+1) + \sum_{r=1}^{5}(-4r) + \sum_{r=1}^{15}(r) + \sum_{r=1}^{20}(-2r+1)$$

$= = 10(11) + 10 + (-2)(5)(6) + 15(8) - 400$

$= -220$

Q. **If A is an involutory matrix given by** $A=\begin{bmatrix}0 & 1 & -1\\ 4 & -3 & 4\\ 3 & -3 & 4\end{bmatrix}$, **then the inverse of** $\frac{1}{2}A$ **will be**

(A) 2A **(B)** $\frac{1}{2}A^{-1}$ **(C)** $\frac{1}{2}A$ **(D)** A^2

Sol. **(A)**

It is given that A is an involutory matrix.

$\therefore A^2 = I$

$$\Rightarrow AA = I \Rightarrow \left(\frac{1}{2}A\right)(2A) = I \Rightarrow \left(\frac{1}{2}A\right)^{-1} = (2A)$$

ADJOINT OF A SQUARE MATRIX

If every element of matrix A is replaced by their corresponding co-factors and transpose of matrix thus obtained will be required adjoint of given matrix A.

- Thus, adj $A = [C_{ij}]^T$ or $(\text{adj } A)_{ij} = C_{ji}$
- If $A=\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}$, then
- $\text{adj } A=\begin{bmatrix}C_{11} & C_{12} & C_{13}\\ C_{21} & C_{22} & C_{23}\\ C_{31} & C_{32} & C_{33}\end{bmatrix}^T=\begin{bmatrix}C_{11} & C_{21} & C_{31}\\ C_{12} & C_{22} & C_{32}\\ C_{13} & C_{23} & C_{33}\end{bmatrix}$

where C_{ij} denotes the cofactor of a_{ij} in A.

Example :

$A=\begin{bmatrix}p & q\\ r & s\end{bmatrix}$

Here, $C_{11} = s$, $C_{12} = -r$, $C_{21} = -q$, $C_{22} = p$. Therefore,

$\text{adj } A=\begin{bmatrix}s & -r\\ -q & p\end{bmatrix}^T=\begin{bmatrix}s & -q\\ -r & p\end{bmatrix}$

Inverse of A Matrix

A non-singular square matrix of order n is invertible if there exists a square matrix B of the same order such that $AB = I_n = BA$

In such a case, we say that the inverse of A is B and we write $A^{-1} = B$

Also from $A(\text{adj } A) = |A| I_n = (\text{adj } A) A$, we can conclude that $A^{-1} = \frac{1}{|A|} \text{adj} A$

Properties of Adjoint and Inverse of Matrix

(i) Let A be a square matrix of order n.

Then $A(\text{adj } A) = |A| I_n = (\text{adj } A) A$.

Proof :

$$(A(\text{adj } A))_{ij} = \sum_{r=1}^{n} (A)_{ir} (\text{adj } A)_{rj} = \sum_{r=1}^{n} a_{ir} C_{jr} = \begin{cases} |A| & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

$$\Rightarrow A(\text{adj } A) = \begin{bmatrix} |A| & 0 & 0 & \dots & 0 \\ 0 & |A| & 0 & \dots & 0 \\ 0 & 0 & |A| & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & |A| \end{bmatrix} = |A| I_n$$

Similarly, $((\text{adj } A)A)_{ij} = \sum_{r=1}^{n} (\text{adj } A)_{ir} (A)_{rj} = \sum_{r=1}^{n} C_{ri} a_{rj} = \begin{cases} |A| & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$

Hence, $A(\text{adj } A) = |A| I_n = (\text{adj } A)A$

(ii) Every invertible matrix possesses a unique inverse.

(iii) Reversal law If A and B are invertible matrices of the same order, then AB is invertible and $(AB)^{-1} = B^{-1} A^{-1}$. In general, if A, B, C, ... are invertible Matrices, then $(ABC\cdots)^{-1} = \cdots C^{-1} B^{-1} A^{-1}$.

(iv) If A is an invertible square matrix, then A^T is also invertible and $(A^T)^{-1} = (A^{-1})^T$.

(v) If A is a non-singular square matrix of order n, then $|\text{adj } A| = |A|^{n-1}$.

(vi) Reversal law for adjoint : If A and B are non-singular square matrices of the same order, then adj (AB) = (adj B) (adj A) (using $(AB)^{-1} = B^{-1} \cdot A^{-1}$) adj (ABC) = (adj C) (adj B) (adj A)

(vii) If A is an invertible square matrix, then

$\text{adj } (A^T) = (\text{adj } A)^T$ (using $(A^T)^{-1} = (A^{-1})^T$)

(viii) If A is a non-singular square matrix, then $\text{adj } (\text{adj } A) = |A|^{n-2} A$

(ix) If A is a non-singular matrix, then $|A^{-1}| = |A|^{-1}$, i.e., $|A^{-1}| = 1/|A|$.

(x) Inverse of the kth power of A is the kth power of the inverse of A.

Q. **Find the adjoint of matrix.** $A=\begin{bmatrix}1 & 1 & 1\\ 2 & 1 & -3\\ -1 & 2 & 3\end{bmatrix}$

Sol. $A=\begin{bmatrix}1 & 1 & 1\\ 2 & 1 & -3\\ -1 & 2 & 3\end{bmatrix}$

$M_{11}=\begin{vmatrix}1 & -3\\ 2 & 3\end{vmatrix}=9 \Rightarrow C_{11}=9$

$M_{12}=\begin{vmatrix}2 & -3\\ -1 & 3\end{vmatrix}=3 \Rightarrow C_{12}=-3$

$M_{13}=\begin{vmatrix}2 & 1\\ -1 & 2\end{vmatrix}=5 \Rightarrow C_{13}=5$

$M_{21}=\begin{vmatrix}1 & 1\\ 2 & 3\end{vmatrix}=1 \Rightarrow C_{11}=-1$

Similarly finding all elements of cofactor matrix, we get:

$$\text{Cofactor (A)}=\begin{bmatrix}9 & -3 & 5\\ -1 & 4 & -3\\ -4 & 5 & -1\end{bmatrix}$$

$$\text{Adj(A)} = (\text{cofactor A})^T=\begin{bmatrix}9 & -1 & -4\\ -3 & 4 & 5\\ 5 & -3 & -1\end{bmatrix}$$

Q. **If** $A=\begin{bmatrix}2 & -3\\ -4 & 1\end{bmatrix}$, **then** $\text{adj}\left(3A^2+12A\right)$ **is equal to**

(A) $\begin{bmatrix}51 & 63\\ 84 & 72\end{bmatrix}$ **(B)** $\begin{bmatrix}51 & 84\\ 63 & 72\end{bmatrix}$ **(C)** $\begin{bmatrix}72 & -63\\ -84 & 51\end{bmatrix}$ **(D)** $\begin{bmatrix}72 & -84\\ -63 & 51\end{bmatrix}$

Sol. **(A)**

Given, $A=\begin{bmatrix}2 & -3\\ -4 & 1\end{bmatrix}$

Then, A satisfies the characteristic equation $A^2-3A-10I=0$

Now $3A^2+12A=3(3A+10I)+12A=21A+30I$

$$=\begin{bmatrix}42 & -63\\ -84 & 21\end{bmatrix}+\begin{bmatrix}30 & 0\\ 0 & 30\end{bmatrix}=\begin{bmatrix}72 & -63\\ -84 & 51\end{bmatrix}$$

$$\therefore \text{adj}\left(3A^2+12A\right)=\begin{bmatrix}51 & 63\\ 84 & 72\end{bmatrix}$$

Q. **Find the matrix A satisfying the matrix equation** $\begin{bmatrix}2 & 1\\3 & 2\end{bmatrix}A\begin{bmatrix}-3 & 2\\5 & -3\end{bmatrix}=\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$

Sol. Let $B=\begin{bmatrix}2 & 1\\3 & 2\end{bmatrix}$ and $C=\begin{bmatrix}-3 & 2\\5 & -3\end{bmatrix}$

$BAC = I$

$\Rightarrow$ $Adj(B)\ BAC\ (Adj\ (C)) = Adj(B)\ I(Adj(C))$

$\Rightarrow$ $(|B|\ I)\ A(|C|I) = Adj(B)\ Adj(C)$

$\Rightarrow$ $|B|\ |C|\ A = Adj\ (B)\ Adj\ (C)$

$\Rightarrow$ $A=\dfrac{1}{1\times(-1)}\begin{bmatrix}2 & -1\\-3 & 2\end{bmatrix}\begin{bmatrix}-3 & -2\\-5 & -3\end{bmatrix}$

$A=\begin{bmatrix}1 & 1\\1 & 0\end{bmatrix}$

Q. **Let** $A=\begin{bmatrix}1 & -1 & 1\\2 & 1 & -3\\1 & 1 & 1\end{bmatrix}$ **and** $10B=\begin{bmatrix}4 & 2 & 2\\-5 & 0 & \alpha\\1 & -2 & 3\end{bmatrix}$ **If B is the inverse of A, then find the value of** α.

(A) 5 **(B) 7** **(C) 9** **(D) 11**

Sol. **(B)**

As B is inverse of A

$AB = I;\ 10AB = 10I$

$$\begin{bmatrix}1 & -1 & 1\\2 & 1 & -3\\1 & 1 & 1\end{bmatrix}\begin{bmatrix}4 & 2 & 2\\-5 & 0 & \alpha\\1 & -2 & 3\end{bmatrix}=10\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$$

$$\begin{bmatrix}10 & 0 & 5-\alpha\\0 & 10 & \alpha-5\\0 & 0 & \alpha+5\end{bmatrix}=\begin{bmatrix}10 & 0 & 0\\0 & 10 & 0\\0 & 0 & 10\end{bmatrix}$$

$\alpha = 5$

Q. **Find the matrix A satisfying the matrix equation** $\begin{bmatrix}1 & 2\\2 & 3\end{bmatrix}A\begin{bmatrix}4 & 7\\3 & 5\end{bmatrix}=\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$.

(A) $\begin{bmatrix}21 & -29\\-13 & 18\end{bmatrix}$ **(B)** $\begin{bmatrix}-21 & 29\\13 & -18\end{bmatrix}$ **(C)** $\begin{bmatrix}21 & 29\\13 & 18\end{bmatrix}$ **(D)** $\begin{bmatrix}-21 & -29\\-13 & -18\end{bmatrix}$

Sol. **(A)**

Let $B=\begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}$ and $C=\begin{bmatrix}4 & 7\\ 3 & 5\end{bmatrix}$, then the given matrix equation becomes BAC=I

Now, $|B| = 3 - 4 = -1 \neq 0$

and $|C| = 20 - 21 = -1 \neq 0$

So, B^{-1} and C^{-1} exist.

$\therefore \quad B^{-1}(BAC)C^{-1} = B^{-1}/C^{-1}$

$\Rightarrow \quad |A| = B^{-1}C^{-1}$

$\Rightarrow \quad A = B^{-1}C^{-1}$ ...(i)

Now, $\text{adj } B=\begin{bmatrix}3 & -2\\ -2 & 1\end{bmatrix}$

$$\therefore B^{-1}=\frac{1}{|B|}(\text{adj}\,B)=\frac{1}{-1}\begin{bmatrix}3 & -2\\ -2 & 1\end{bmatrix}=\begin{bmatrix}-3 & 2\\ 2 & -1\end{bmatrix}$$

Also, $\text{adj } C=\begin{bmatrix}5 & -7\\ -3 & 4\end{bmatrix}$

$$\therefore C^{-1}=\frac{1}{|C|}(\text{adj}\,C)=\frac{1}{-1}\begin{bmatrix}5 & -7\\ -3 & 4\end{bmatrix}=\begin{bmatrix}-5 & 7\\ 3 & -4\end{bmatrix}$$

Now, from (i), $A=B^{-1}C^{-1}=\begin{bmatrix}-3 & 2\\ 2 & -1\end{bmatrix}\begin{bmatrix}-5 & 7\\ 3 & -4\end{bmatrix}$

$$=\begin{bmatrix}15+6 & -21-8\\ -10-3 & 14+4\end{bmatrix}=\begin{bmatrix}21 & -29\\ -13 & 18\end{bmatrix}$$

Q. **Let** $a=\lim\limits_{x\to 1}\left(\frac{x}{\ln x}-\frac{1}{x\ln x}\right)$; $b=\lim\limits_{x\to 0}\frac{x^3-16x}{4x+x^2}$; $c=\lim\limits_{x\to 0}\frac{\ln(1+\sin x)}{x}$ **and**

$d=\lim\limits_{x\to -1}\frac{(x+1)^3}{3(\sin(x+1)-(x+1))}$, **then the matrix** $\begin{bmatrix}a & b\\ c & d\end{bmatrix}$ **is**

(A) Idempotent **(B) Involutory**

(C) Identity **(D) Nilpotent**

Sol. **(D)**

$a = 2; b = -4; c = 1; d = -2$

Let $A=\begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix}$, then $A^2=\begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix}\begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$ = null matrix, hence A is nilpotent.

Note that any matrix of the form $\begin{bmatrix}a & -a^2\\ 1 & -a\end{bmatrix}$ is a nilpotent.

Q. If $A = \begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix}$, then sum of all entries of the matrix $B = (I - A)(I + A)^{-1}$

(A) $-\frac{34}{13}$ **(B)** $-\frac{14}{13}$ **(C)** $-\frac{12}{13}$ **(D)** $-\frac{24}{13}$

Sol. **(D)**

$B = (I - A)(I + A)^{-1}$ for $A = \begin{bmatrix} 0 & 5 \\ -5 & 0 \end{bmatrix}$

$$= \begin{bmatrix} 1 & -5 \\ 5 & 1 \end{bmatrix}\begin{bmatrix} 1 & 5 \\ -5 & 1 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} 1 & -5 \\ 5 & 1 \end{bmatrix}\left(\frac{1}{26}\right)\begin{bmatrix} 1 & -5 \\ 5 & 1 \end{bmatrix} = \left(\frac{1}{26}\right)\begin{bmatrix} 1 & -5 \\ 5 & 1 \end{bmatrix}\begin{bmatrix} 1 & -5 \\ 5 & 1 \end{bmatrix}$$

$$= \frac{1}{26}\begin{bmatrix} -24 & -10 \\ 10 & -24 \end{bmatrix} = \frac{1}{13}\begin{bmatrix} -12 & -5 \\ 5 & -12 \end{bmatrix}$$

Previous Year's Question

Let M and N be two 3 × 3 non-singular skew symmetric matrices such that MN = NM. If P^T denotes the transpose of P, then $M^2N^2(M^TN)^{-1}(MN^{-1})^T$ is equal to **[JEE 2011]**

(A) M^2 (B) $-N^2$ (C) $-M^2$ (D) MN

Sol. **(C)**

$\because$ MN = NM

$M^2N^2 = MN\ MN$

Given, $M^2N^2(M^TN)^{-1} \cdot (MN^{-1})^T$

$= -\ MN\ MN\ N^{-1}M^{-1}\ N^{-1}\ M$ $\quad \because (M^T)^{-1} = (-M)^{-1} = -\ M^{-1}$

$= -\ M\ NN^{-1}M = -\ M^2$

The most appropriate answer to this question is (C), But given information is contradictory as skew symmetric matrix of odd order cannot be non - singular. IIT has declared this question as bonus (Marks to all students)

Previous Year's Question

If the adjoint of a 3 × 3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible value(s) of the determinant of P is(are) **[JEE 2012]**

(A) –2 (B) –1 (C) 1 (D) 2

Sol. **(A,D)**

$$\text{adj } (P) = \begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$$

$|\text{adj } P| = (3-7) - 4(6-7) + 4(2-1) = 4$

$|\text{adj } P| = |P|^{n-1} = |P|^2$

$|\text{adj } P| = 4 = |P|^2$

$|P| = 2 \text{ or } -2$

System of Linear Equations in Three Unknowns

Consider the system of equations

$a_1x + b_1y + c_1z = d_1$ ...(i)

$a_2x + b_2y + c_2z = d_2$ ...(ii)

$a_3x + b_3y + c_3z = d_3$ ...(iii)

Methods of solving non-homogenous equation in three variables:

(a) Determinant method (Cramer' s Rule)

which has already been dealt with in the chapter on Determinants.

(b) Matrix method

The given system of equations can be written as:

$$AX = B \text{ where } A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

Criteria for Consistency

(i) If $|A| \neq 0$, System is consistent having unique solution

(a) If $|A| \neq 0$, and (adj A). B ≠ Null matrix

System is consistent having unique non-trivial solution

(b) If $|A| \neq 0$, and (adj A). B = Null matrix
System is consistent having unique trivial solution

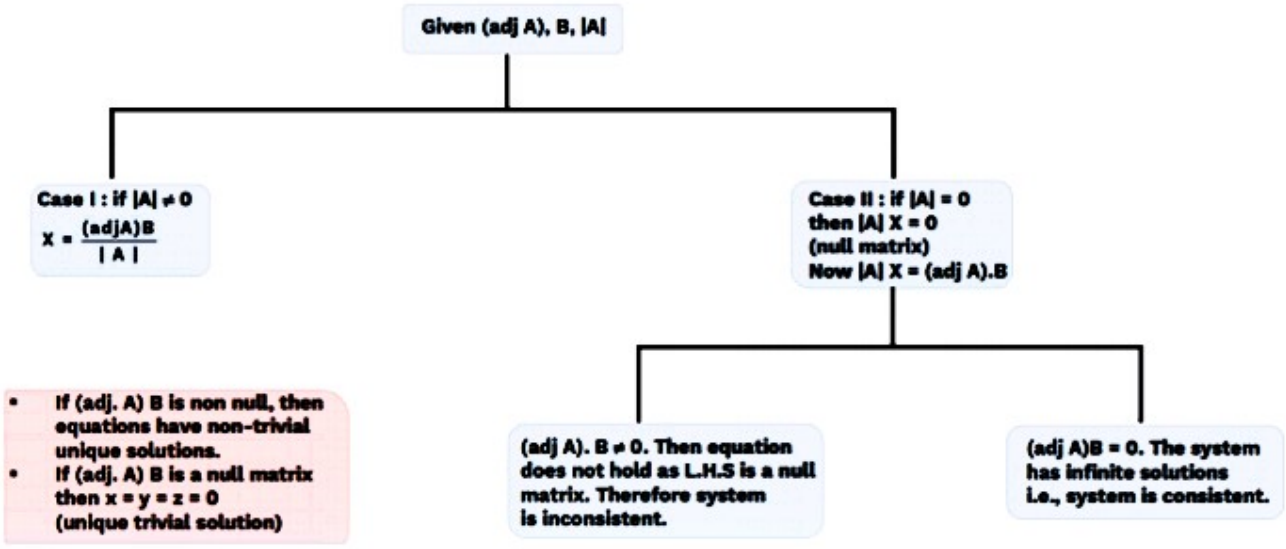
(ii) If $|A| = 0$, then the system of equation given by AX = B can be consistent with infinitely many solution or it can be inconsistent also.

Case I : If (adj A). B = Null matrix
$\Rightarrow$ Infinitely many solutions (System Consistent)

Case II : If (adj A). B $\neq$ Null matrix $\Rightarrow$ No solution (System Inconsistent)

Example :

$2x + y = 3$ and $4x + 2y = 6$
$\Rightarrow |A| = 0$ so A^{-1} does not exist, but system has infinitely many solutions.
i.e., $x = m$ and $y = 3 - 2m$ is solution.



Q. **Let S be the set of all real values of k for which the system of linear equations $x + y + z = 2$; $2x + y - z = 3$; $3x + 2y + kz = 4$ has a unique solution. Then S is**

(A) an empty set **(B) equal to R**
(C) equal to {0} **(D) equal to R – {0}**

Sol. **(D)**

For unique solution, $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & k \end{vmatrix} \neq 0$

$\Rightarrow 1(k + 2) - 1(2k + 3) + 1(4 - 3) \neq 0$
$\Rightarrow -k + 2 - 3 + 1 \neq 0$
$\Rightarrow k \neq 0$ $\quad \therefore S = R - \{0\}$

Q. **If the system of linear equations : $x + ay + z = 3$ $x + 2y + 2z = 6$ $x + 5y + 3z = b$ has no solution, then**

(A) $a = -1, b = 9$ **(B) $a \neq -1, b = 9$** **(C) $a = 1, b \neq 9$** **(D) $a = -1, b \neq 9$**

Sol. **(D)**

The given system of equations has no solution.

$$\therefore \begin{vmatrix} 1 & a & 1 \\ 1 & 2 & 2 \\ 1 & 5 & 3 \end{vmatrix} = 0$$

Applying $R_1 \to R_1 - R_2$ and then $R_2 \to R_2 - R_3$, we get

$$\begin{vmatrix} 0 & a-2 & -1 \\ 0 & -3 & -1 \\ 1 & 5 & 3 \end{vmatrix} = 0 \Rightarrow a = -1$$

Also in addition to above (adj A) $B \neq O \Rightarrow$ no solution.

$$\text{Here, } \text{adj}(A) = \begin{bmatrix} -4 & 8 & -4 \\ -1 & 2 & -1 \\ 3 & -6 & 3 \end{bmatrix}$$

$$\Rightarrow (\text{adj}\, A)(B) = \begin{bmatrix} -4 & 8 & -4 \\ -1 & 2 & -1 \\ 3 & -6 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ b \end{bmatrix} \neq O$$

$$\Rightarrow -12 + 48 - 4b \neq 0 \Rightarrow b \neq 9$$

Q. **Consider the system of linear equations**

$x_1 + 2x_2 + x_3 = 3$ $2x_1 + 3x_2 + x_3 = 3$

$3x_1 + 5x_2 + 2x_3 = 1$

The system has

(A) infinite number of solutions **(B) exactly 3 solutions**

(C) a unique solution **(D) no solution**

Sol. **(D)**

Given system of equations is

$x_1 + 2x_2 + x_3 = 3$ $2x_1 + 3x_2 + x_3 = 3$

$3x_1 + 5x_2 + 2x_3 = 1$

It can be observed that the sum of first two equations yields

$$(x_1 + 2x_2 + x_3) + (2x_1 + 3x_2 + x_3) = 3 + 3$$

$$\Rightarrow 3x_1 + 5x_2 + 2x_3 = 6$$

But this contradicts the third equation, i.e., $3x_1 + 5x_2 + 2x_3 = 1$

So, such the system is inconsistent and hence it has no solution.

Q. **The set of all values of λ for which the system of linear equations**

$2x_1 - 2x_2 + x_3 = \lambda x_1$

$2x_1 - 3x_2 + 2x_3 = \lambda x_2$

$-x_1 + 2x_2 = \lambda x_3$

has a non-trivial solution,

(A) contains two elements. **(B) contains more than two elements.**

(C) is an empty set. **(D) is a singleton set.**

Sol. **(A)**

The given system is

$(2-\lambda)x_1 - 2x_2 + x_3 = 0$

$2x_1 - (3+\lambda)x_2 + 2x_3 = 0$

$-x_1 + 2x_2 - \lambda x_3 = 0$

For non-trivial solution, the determinant of the coefficient matrix must be zero.

$$\begin{vmatrix} 2-\lambda & -2 & 1 \\ 2 & -3-\lambda & 2 \\ -1 & 2 & -\lambda \end{vmatrix} = 0$$

Therefore

$\Rightarrow (2-\lambda)\{\lambda(3+\lambda)-4\} + 2\{-2\lambda+2\} + 1\{4-(3+\lambda)\} = 0$

$\Rightarrow (2-\lambda)(\lambda^2+3\lambda-4) - 4\lambda + 4 + 1 - \lambda = 0$

$\Rightarrow (2-\lambda)(\lambda^2+3\lambda-4) - 5\lambda + 5 = 0$

$\Rightarrow (2-\lambda)(\lambda-1)(\lambda+4) - 5(\lambda-1) = 0$

$\Rightarrow (\lambda-1)(-\lambda^2-2\lambda+8-5) = 0$

$\Rightarrow (\lambda-1)(\lambda^2+2\lambda-3) = 0$

$\Rightarrow (\lambda-1)^2(\lambda+3) = 0$

Thus, $\lambda = 1, 1, -3$

$\therefore$ Set of values of λ contain 2 elements.

Previous Year's Question

The number of 3 × 3 matrices A whose entries are either 0 or 1 and for which the system $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has exactly two distinct solutions, is **[JEE 2010]**

(A) 0 (B) $2^9 - 1$ (C) 168 (D) 2

Sol. Let $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$. The system becomes

$a_1x + b_1y + c_1z = 0$,
$a_2x + b_2y + c_2z = 0$ and $a_3x + b_3y + c_3z = 0$ representing 3 planes.

Three planes meets at

(i) exactly one point, or
(ii) no point
(iii) infinite number of points
They can't meet at exactly two points for any A.

Q. **If the system of equations**
$x - 2y + z = a$
$2x + y - 2z = b$
and $x + 3y - 3z = c$
have at least one solution, then
(A) $a + b + c = 0$ **(B) $a - b + c = 0$**
(C) $-a + b + c = 0$ **(D) $a + b - c = 0$**

Sol. **(B)**

$$\Delta = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -2 \\ 1 & 3 & -3 \end{vmatrix} = -3 + 6 + 2(-6 + 2) + (6 - 1) = 0$$

Hence for atleast one solution
if $\Delta_1 = \Delta_2 = \Delta_3 = 0$

$$\therefore \quad \Delta_1 = \begin{vmatrix} a & -2 & 1 \\ b & 1 & -2 \\ c & 3 & -3 \end{vmatrix} = 0$$

$\Rightarrow \quad a - b + c = 0$

From $\Delta_2 = 0$ and $\Delta_3 = 0$, we get the same condition.

Q. **If A, B, C are the angles of a triangle, the system of equations $(\sin A)x + y + z = \cos A$, $x + (\sin B)y + z = \cos B$, $x + y + (\sin C)z = 1 - \cos C$ has**
(A) No solution **(B) Unique solution**
(C) Infinitely many solutions **(D) Finitely many solutions**

Sol. **(B)**

$$\text{Let } \Delta = \begin{vmatrix} \sin A & 1 & 1 \\ 1 & \sin B & 1 \\ 1 & 1 & \sin C \end{vmatrix}$$

Apply $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$

$$\Delta = \begin{vmatrix} \sin A & 1-\sin A & 1-\sin A \\ 1 & \sin B - 1 & 0 \\ 1 & 0 & \sin C - 1 \end{vmatrix}$$

Then expanded along C_1, we get

$\Delta = \sin A(1 - \sin B)(1 - \sin C) + (1 - \sin A)(1 - \sin C) + (1 - \sin A)(1 - \sin B)$

Since, A, B, C are angles of a triangle, $0 < \sin A, \sin B, \sin C \leq 1$

$\Rightarrow \quad \Delta \neq 0 \Rightarrow$ Unique solution

Cayley - Hamilton Theorem :

Let A be a square matrix, then the equation $|A - xI| = 0$ is called characteristic equation of A. After solving the characteristic equation the values of 'x' are said to be characteristic roots of the equation.

Note :

(i) Sum of the roots of the characteristic equation is equal to trace of the matrix.

(ii) Product of the roots of the characteristic equation is equal to the determinant value.

(iii) The degree of characteristic equation is same as the order of the matrix.

Note :

This theorem is helpful to find the inverse of any non-singular square matrix.

i.e. $a_0A^n + a_1A^{n-1} + \ldots\ldots + a_{n-1}A + a_nI = O$

On multiplying by A^{-1} on both the sides of above equation, we get

$$A^{-1} = -\frac{1}{a_n}\left(a_0A^{n-1} + a_1A^{n-2} + \ldots\ldots a_{n-1}I\right)$$

Point to Remember!!!

Every square matrix A satisfy its characteristic equation

i.e. $a_0x^n + a_1x^{n-1} + \ldots\ldots + a_{n-1}x + a_n = 0$ is the characteristic equation of A, then

$a_0A^n + a_1A^{n-1} + \ldots\ldots + a_{n-1}A + a_nI = O$

Q. Let $A = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$ be a square matrix and $A^4 = = eA + mI$, then tell the value of e

(A) 5 **(B) 7** **(C) 9** **(D) 11**

Sol. **(C)**

$$A - \lambda I = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & 1 \\ -1 & 1-\lambda \end{bmatrix}$$

Now, $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 1 \\ -1 & 1-\lambda \end{vmatrix} = 0$$

$\Rightarrow (2 - \lambda)(1 - \lambda) + 1 = 0$

$\Rightarrow \lambda^2 - 3\lambda + 3 = 0$

$\Rightarrow A^2 - 3A + 3I = 0$

$\Rightarrow A^2 = 3(A - I)$

$\Rightarrow A^4 = 9(A^2 - 2A + I)$

$= 9(3A - 3I - 2A + I)$

$= 9(A - 2I) = 9A - 18I \Rightarrow e = 9, m = -18$

Q. $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ If $A^{-1} = kA^2 + xA + rI$. Tell the value of $5(k + x) - r$

(A) 5 **(B) 15** **(C) 9** **(D) 3**

Sol. **(D)**

$$A - \lambda I = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 & 0 \\ 2 & -1-\lambda & 0 \\ 0 & 0 & -1-\lambda \end{bmatrix}$$

$|A - \lambda I| = (1 - \lambda)\{(-1 - \lambda)^2 - 0\} - 2\{-2(1 + \lambda)\} = -(\lambda - 1)(\lambda + 1)^2 + 4(\lambda + 1)$

$|A - \lambda I| = 0 \Rightarrow (\lambda - 1)(\lambda + 1)^2 - 4(\lambda + 1) = 0 \Rightarrow \lambda^3 + \lambda^2 - \lambda - 1 - 4\lambda - 4 = 0$

$\Rightarrow \lambda^3 + \lambda^2 - 5\lambda - 5 = 0$

So, characteristic equation $A^3 + A^2 - 5A - 5I = 0$

$$\Rightarrow A^{-1} = \frac{A^2}{5} + \frac{A}{5} - I \Rightarrow k = \frac{1}{5}, x = \frac{1}{5}, r = -1 \Rightarrow 5(k + x) - r = 3$$

Q. If $\begin{bmatrix} 1 & 2 & a \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & 18 & 2007 \\ 0 & 1 & 36 \\ 0 & 0 & 1 \end{bmatrix}$, then the value of a is:

(A) 205 (B) 107 (C) 191 (D) 110

Sol. (C)

$$A = \begin{bmatrix} 1 & 2 & a \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^2 = AA = \begin{bmatrix} 1 & 4 & 2a+8 \\ 0 & 1 & 8 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^3 = A^2A = \begin{bmatrix} 1 & 6 & 3a+24 \\ 0 & 1 & 12 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^4 = A^3A = \begin{bmatrix} 1 & 8 & 4a+48 \\ 0 & 1 & 16 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^n = \begin{bmatrix} 1 & n^{th}\text{term}(2,4,6,....) & n^{th}\text{term}(a,2a+8,3a+24...) \\ 0 & 1 & n^{th}\text{term}(4,8,12...) \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 18 & 2007 \\ 0 & 1 & 36 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2n & n^{th}\text{term}(a,2a+8,3a+24...) \\ 0 & 1 & 4n \\ 0 & 0 & 1 \end{bmatrix}$$

$$S = a + (2a+8) + (3a+24) + (4a+48) + ... + T_n$$

$$\begin{array}{lllllll} S = & a & +(2a+8) & +(3a+24)+... & & +T_n \\ - & - & - & - & - & - \\ \hline 0 = & a & +(a+8) & +(a+16)+(a+24)+... & & -T_n \end{array}$$

$$T_n = na + \frac{n-1}{2}[2\times 8 + (n-1-1)8]$$

$$= na + (n-1)[4n]$$

$$= na + 4n^2 - 4n$$

$$\begin{bmatrix} 1 & 2n & na + 4n^2 - 4n \\ 0 & 1 & 4n \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 18 & 2007 \\ 0 & 1 & 36 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow$ $2n = 18,\ na + 4n^2 - 4n = 2007$

$\Rightarrow$ $n = 9,\ 4n^2 - 4n + na = 2007$

$\Rightarrow$ $9a + 4 \times 9 \times 8 = 2007$

$\Rightarrow$ $a + 32 = 223$

$\Rightarrow$ $a = 191$